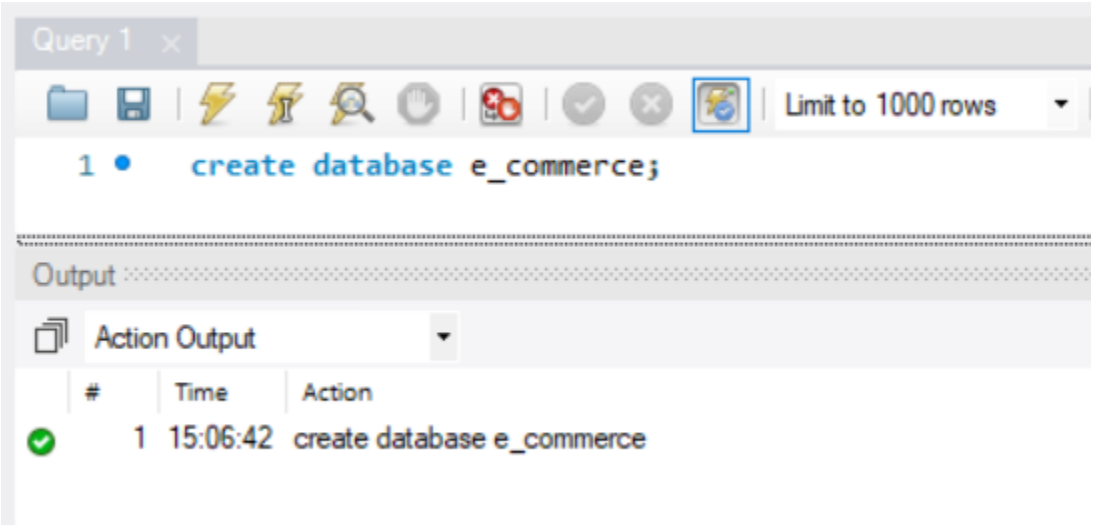
Kshitiz Gupta

SQL Assignment : E-Commerce Database

1. Database Creation

Query: create database e_commerce;

Screenshot:



2. Table Creation

Customers Table Query:

```
CREATE TABLE Customers (
   customer_id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(50),
   email VARCHAR(50),
   mobile VARCHAR(15)
);
```

Screenshot:

```
use e_commerce;
CREATE TABLE Customers (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    email VARCHAR(50),
    mobile VARCHAR(15)
);
```

Output

Action Output

#	Time	Action	Message
1	15:10:05	use e_commerce	0 row(s) affected
2	15:10:38	CREATE TABLE Customers (customer_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50), ...	0 row(s) affected

Products Table Query:

```
CREATE TABLE Products (
   id INT,
   name VARCHAR(50) NOT NULL,
   description VARCHAR(200),
   price DECIMAL(10,2) NOT NULL,
   category VARCHAR(50)
);
```

Screenshot:

```
CREATE TABLE Products (
    id INT,
    name VARCHAR(50) NOT NULL,
    description VARCHAR(200),
    price DECIMAL(10,2) NOT NULL,
    category VARCHAR(50)
);
```

Output

Action Output

#	Time	Action	Message
1	15:14:37	CREATE TABLE Products (id INT, name VARCHAR(50) NOT NULL, description VARCHAR(200), pric...	0 row(s) affected

3. Table Modification (ALTER)

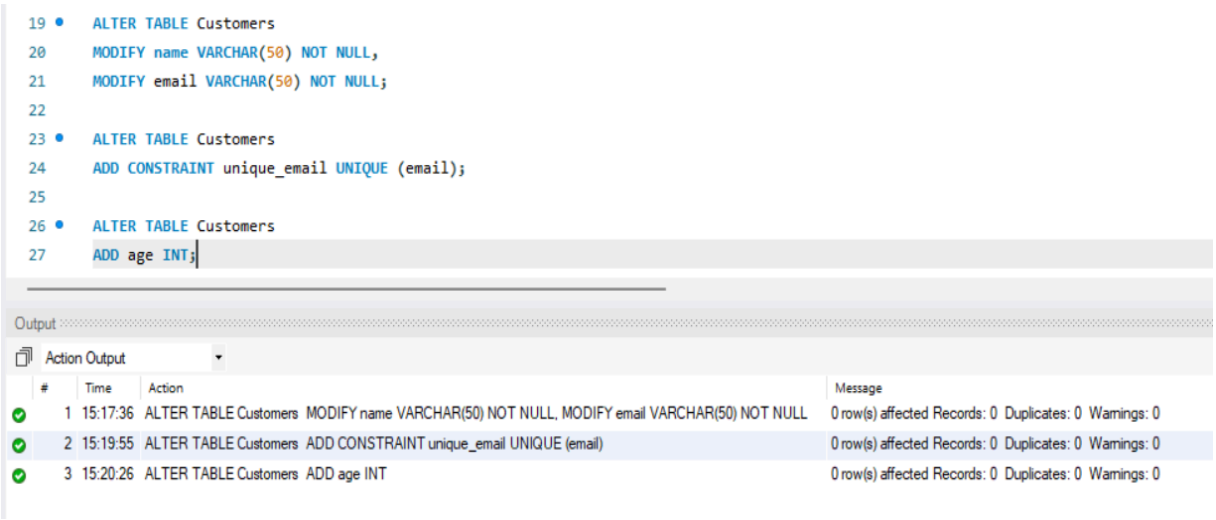
Customers Table Modifications:

ALTER TABLE Customers
MODIFY name VARCHAR(50) NOT NULL,
MODIFY email VARCHAR(50) NOT NULL;

ALTER TABLE Customers
ADD CONSTRAINT unique_email UNIQUE (email);

ALTER TABLE Customers
ADD age INT;

Screenshot:



Products Table Modifications:

ALTER TABLE Products
CHANGE id product_id INT;

ALTER TABLE Products
ADD PRIMARY KEY (product_id);

ALTER TABLE Products
MODIFY product_id INT AUTO_INCREMENT;

ALTER TABLE Products
MODIFY description TEXT;

Screenshot:

```
ALTER TABLE Products
CHANGE id product_id INT;

ALTER TABLE Products
ADD PRIMARY KEY (product_id);

ALTER TABLE Products
MODIFY product_id INT AUTO_INCREMENT;

ALTER TABLE Products
MODIFY description TEXT;
```

Output

Action Output

#	Time	Action	Message
1	15:22:23	ALTER TABLE Products CHANGE id product_id INT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
2	15:23:48	ALTER TABLE Products ADD PRIMARY KEY (product_id)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
3	15:23:54	ALTER TABLE Products ADD PRIMARY KEY (product_id)	Error Code: 1068. Multiple primary key defined
4	15:24:16	ALTER TABLE Products MODIFY product_id INT AUTO_INCREMENT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
5	15:26:03	ALTER TABLE Products MODIFY description TEXT	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

4. Orders Table Creation & Modification

Query:

```
CREATE TABLE `Order` (
order_id INT AUTO_INCREMENT PRIMARY KEY,
customer_id INT,
product_id INT,
quantity INT NOT NULL,
order_date DATE NOT NULL,
status ENUM('Pending','Success','Cancel'),
payment_method ENUM('Credit','Debit','UPI'),
total_amount DECIMAL(10,2) NOT NULL,
FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

Screenshot:

```
CREATE TABLE `Order` (
    order_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_id INT,
    product_id INT,
    quantity INT NOT NULL,
    order_date DATE NOT NULL,
    status ENUM('Pending','Success','Cancel'),
    payment_method ENUM('Credit','Debit','UPI'),
    total_amount DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)
);
```

Output

Action Output

#	Time	Action	Message
1	15:38:02	CREATE TABLE `Order` (order_id INT AUTO_INCREMENT PRIMARY KEY, customer_id INT, product_id I...	0 row(s) affected

5.Modify Orders Table(using Alter keyword)

Query:

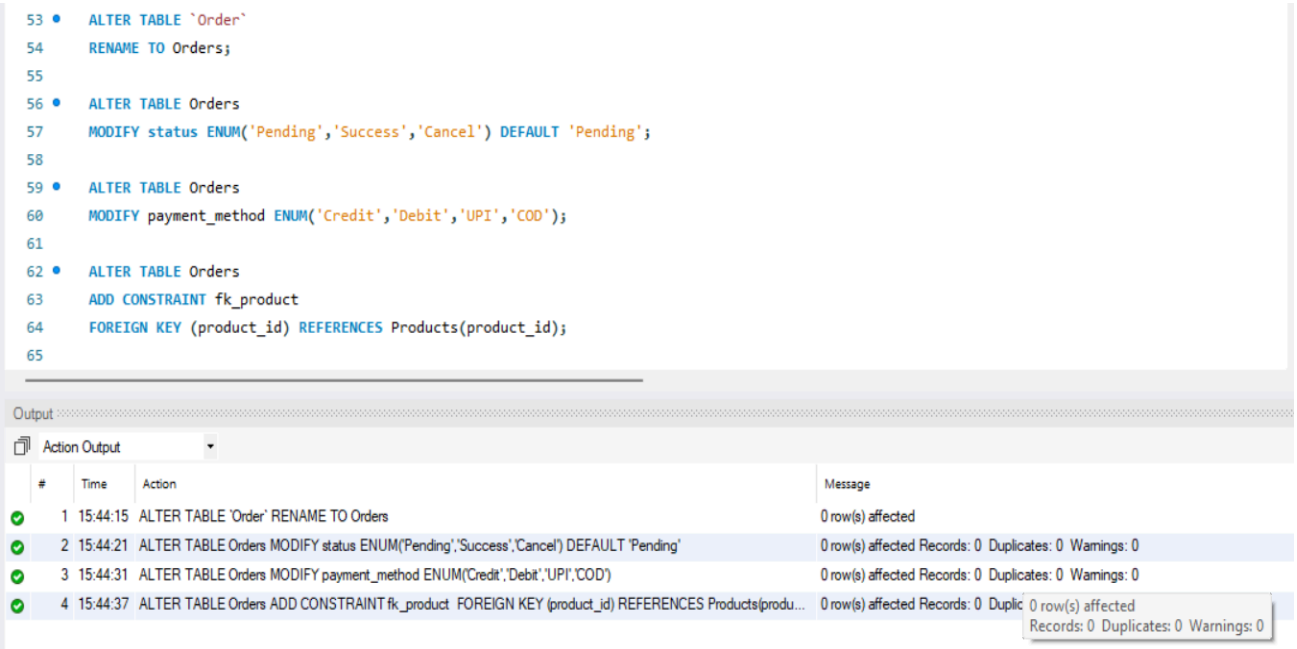
```
ALTER TABLE `Order`
RENAME TO Orders;

ALTER TABLE Orders
MODIFY status ENUM('Pending','Success','Cancel') DEFAULT 'Pending';

ALTER TABLE Orders
MODIFY payment_method ENUM('Credit','Debit','UPI','COD');

ALTER TABLE Orders
ADD CONSTRAINT fk_product
FOREIGN KEY (product_id) REFERENCES Products(product_id);
```

Screenshot:



6. Data Insertion

Note: this is a sample data and not a real data

Customers Table Insert:

INSERT INTO Customers (name, email, mobile, age) VALUES
('Amit Sharma','amit@gmail.com','9876543210',25),
('Riya Singh','riya@gmail.com','9876543211',28),
('Rahul Verma','rahul@gmail.com','9876543212',30),
('Sneha Jain','sneha@gmail.com','9876543213',27),
('Vikas Gupta','vikas@gmail.com','9876543214',35),
('Neha Kapoor','neha@gmail.com','9876543215',26),
('Arjun Mehta','arjun@gmail.com','9876543216',32),
('Pooja Sharma','pooja@gmail.com','9876543217',24),
('Karan Patel','karan@gmail.com','9876543218',29),
('Simran Kaur','simran@gmail.com','9876543219',31),
('Manish Yadav','manish@gmail.com','9876543220',33),
('Anita Das','anita@gmail.com','9876543221',22),
('Rohit Roy','rohit@gmail.com','9876543222',34),
('Meera Nair','meera@gmail.com','9876543223',29),
('Suresh Kumar','suresh@gmail.com','9876543224',40),
('Priya Iyer','priya@gmail.com','9876543225',23),
('Deepak Singh','deepak@gmail.com','9876543226',36),

('Nikita Shah','nikita@gmail.com','9876543227',27),
('Raj Malhotra','raj@gmail.com','9876543228',38),
('Alok Pandey','alok@gmail.com','9876543229',41);

Screenshot:

```
INSERT INTO Customers (name, email, mobile, age) VALUES
('Amit Sharma','amit@gmail.com','9876543210',25),
('Riya Singh','riya@gmail.com','9876543211',28),
('Rahul Verma','rahul@gmail.com','9876543212',30),
('Sneha Jain','sneha@gmail.com','9876543213',27),
('Vikas Gupta','vikas@gmail.com','9876543214',35),
('Neha Kapoor','neha@gmail.com','9876543215',26),
('Arjun Mehta','arjun@gmail.com','9876543216',32),
('Pooja Sharma','pooja@gmail.com','9876543217',24),
('Karan Patel','karan@gmail.com','9876543218',29),
('Simran Kaur','simran@gmail.com','9876543219',31),
('Manish Yadav','manish@gmail.com','9876543220',33),
('Anita Das','anita@gmail.com','9876543221',22),
('Rohit Roy','rohit@gmail.com','9876543222',34),
('Meera Nair','meera@gmail.com','9876543223',29),
('Suresh Kumar','suresh@gmail.com','9876543224',40),
('Priya Iyer','priya@gmail.com','9876543225',23),
('Deepak Singh','deepak@gmail.com','9876543226',36),
('Nikita Shah','nikita@gmail.com','9876543227',27),
('Raj Malhotra','raj@gmail.com','9876543228',38),
('Alok Pandey','alok@gmail.com','9876543229',41);
```

Output

Action Output

#	Time	Action	Message
1	15:52:26	INSERT INTO Customers (name, email, mobile, age) VALUES ('Amit Sharma','amit@gmail.com','9876543210',25), ('Ri...	20 row(s) affected Records: 20 Duplicates: 0 Warnings: 0

Products Table Insert:

```
INSERT INTO Products (name, description, price, category) VALUES
('Laptop','Gaming laptop',800,'Electronics'),
('Mobile','Smartphone',300,'Electronics'),
('Headphones','Wireless',100,'Electronics'),
('Keyboard','Mechanical',70,'Electronics'),
('Mouse','Optical',40,'Electronics'),
('TV','LED TV',600,'Electronics'),
('Tablet','Android tablet',250,'Electronics'),
('Charger','Fast charger',30,'Electronics'),
('Camera','DSLR',900,'Electronics'),
('Speaker','Bluetooth',150,'Electronics'),
('Shoes','Running shoes',120,'Fashion'),
('Shirt','Cotton shirt',40,'Fashion'),
('Jeans','Denim',60,'Fashion'),
('Watch','Analog',200,'Accessories'),
('Bag','Travel bag',80,'Accessories'),
('Bottle','Water bottle',20,'Home'),
('Chair','Office chair',150,'Home'),
('Table','Wooden table',250,'Home'),
('Fan','Ceiling fan',90,'Home'),
('Lamp','Desk lamp',45,'Home');
```

Screenshot:

```sql
INSERT INTO Products (name, description, price, category) VALUES
('Laptop','Gaming laptop',800,'Electronics'),
('Mobile','Smartphone',300,'Electronics'),
('Headphones','Wireless',100,'Electronics'),
('Keyboard','Mechanical',70,'Electronics'),
('Mouse','Optical',40,'Electronics'),
('TV','LED TV',600,'Electronics'),
('Tablet','Android tablet',250,'Electronics'),
('Charger','Fast charger',30,'Electronics'),
('Camera','DSLR',900,'Electronics'),
('Speaker','Bluetooth',150,'Electronics'),
('Shoes','Running shoes',120,'Fashion'),
('Shirt','Cotton shirt',40,'Fashion'),
('Jeans','Denim',60,'Fashion'),
('Watch','Analog',200,'Accessories'),
('Bag','Travel bag',80,'Accessories'),
('Bottle','Water bottle',20,'Home'),
('Chair','Office chair',150,'Home'),
('Table','Wooden table',250,'Home'),
('Fan','Ceiling fan',90,'Home'),
('Lamp','Desk lamp',45,'Home');
```

Output

Action Output

#	Time	Action	Message
1	16:34:56	INSERT INTO Products (name, description, price, category) VALUES ('Laptop','Gaming laptop',800,'Electronics'), ('M...	20 row(s) affected Records: 20 Duplicates: 0 Warnings: 0

Orders Table Insert:

```
INSERT INTO Orders (customer_id, product_id, quantity, order_date, status, payment_method,
total_amount) VALUES
(1,1,1,'2024-01-01','Success','Credit',800),
(2,2,2,'2024-01-02','Pending','Debit',600),
(3,3,1,'2024-01-03','Cancel','UPI',100),
(4,4,3,'2024-01-04','Success','COD',210),
(5,5,2,'2024-01-05','Pending','Credit',80),
(6,6,1,'2024-01-06','Success','Debit',600),
(7,7,2,'2024-01-07','Success','UPI',500),
(8,8,4,'2024-01-08','Pending','COD',120),
(9,9,1,'2024-01-09','Success','Credit',900),
(10,10,2,'2024-01-10','Cancel','Debit',300),
(11,11,1,'2024-01-11','Success','UPI',120),
(12,12,3,'2024-01-12','Pending','COD',120),
(13,13,2,'2024-01-13','Success','Credit',120),
(14,14,1,'2024-01-14','Success','Debit',200),
(15,15,2,'2024-01-15','Pending','UPI',160),
(16,16,5,'2024-01-16','Success','COD',100),
(17,17,1,'2024-01-17','Cancel','Credit',150),
(18,18,2,'2024-01-18','Success','Debit',500),
(19,19,1,'2024-01-19','Pending','UPI',90),
(20,20,3,'2024-01-20','Success','COD',135);
```

Screenshot:

```sql
INSERT INTO Orders (customer_id, product_id, quantity, order_date, status, payment_method, total_amount) VALUES
(1,1,1,'2024-01-01','Success','Credit',800),
(2,2,2,'2024-01-02','Pending','Debit',600),
(3,3,1,'2024-01-03','Cancel','UPI',100),
(4,4,3,'2024-01-04','Success','COD',210),
(5,5,2,'2024-01-05','Pending','Credit',80),
(6,6,1,'2024-01-06','Success','Debit',600),
(7,7,2,'2024-01-07','Success','UPI',500),
(8,8,4,'2024-01-08','Pending','COD',120),
(9,9,1,'2024-01-09','Success','Credit',900),
(10,10,2,'2024-01-10','Cancel','Debit',300),
(11,11,1,'2024-01-11','Success','UPI',120),
(12,12,3,'2024-01-12','Pending','COD',120),
(13,13,2,'2024-01-13','Success','Credit',120),
(14,14,1,'2024-01-14','Success','Debit',200),
(15,15,2,'2024-01-15','Pending','UPI',160),
(16,16,5,'2024-01-16','Success','COD',100),
(17,17,1,'2024-01-17','Cancel','Credit',150),
(18,18,2,'2024-01-18','Success','Debit',500),
(19,19,1,'2024-01-19','Pending','UPI',90),
(20,20,3,'2024-01-20','Success','COD',135);
```

Output

Action Output

#	Time	Action	Message
1	16:34:56	INSERT INTO Products (name, description, price, category) VALUES ('Laptop','Gaming laptop',800,'Electronics'), ('M...	20 row(s) affected Records: 20 Duplicates: 0 Warnings: 0
2	16:37:31	INSERT INTO Orders (customer_id, product_id, quantity, order_date, status, payment_method, total_amount) VALUE...	20 row(s) affected Records: 20 Duplicates: 0 Warnings: 0

7. SQL Queries & Results

Query 1: Count products per category

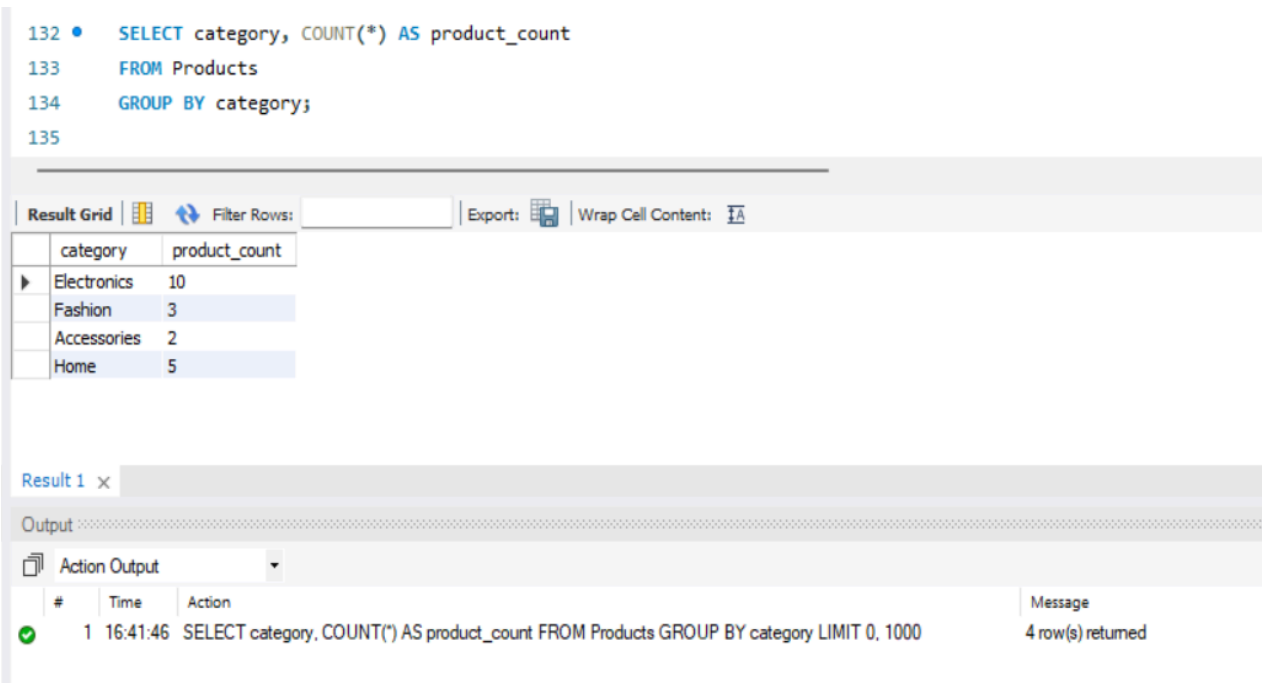
Uses GROUP BY

This query counts total products in each category using GROUP BY.

Query:

```
SELECT category, COUNT(*) AS product_count
FROM Products
GROUP BY category;
```

Screenshot:



Query 2: Electronics filter

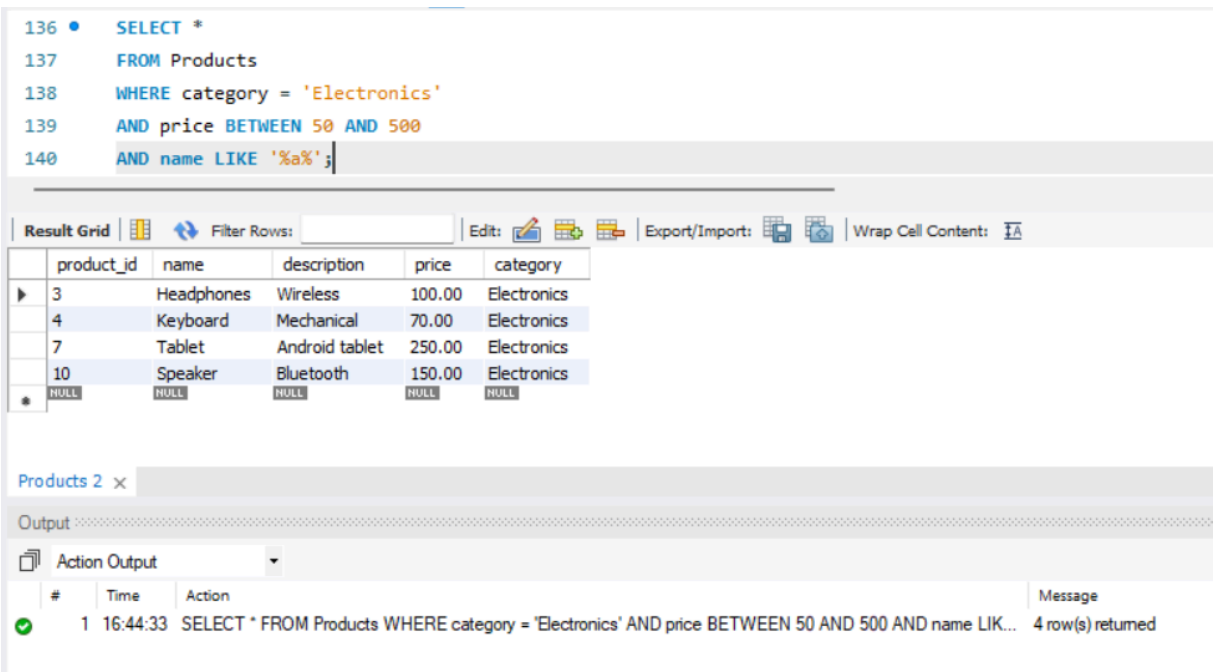
Multiple conditions

This query filters products based on category, price range, and name pattern.

Query:

```
SELECT *
FROM Products
WHERE category = 'Electronics'
AND price BETWEEN 50 AND 500
AND name LIKE '%a%';
```

Screenshot:



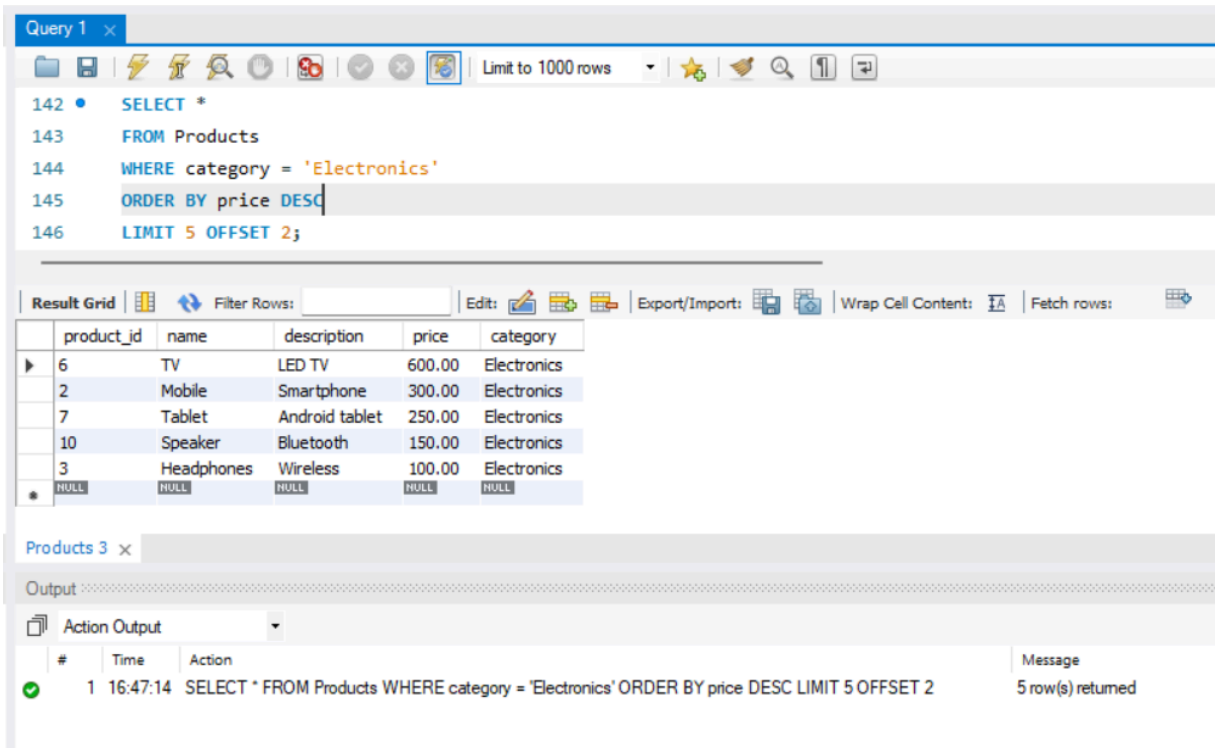
Query 3: Top 5 expensive products (offset 2)

This query retrieves top expensive products while skipping first two using LIMIT and OFFSET.

Query:

```
SELECT *
FROM Products
WHERE category = 'Electronics'
ORDER BY price DESC
LIMIT 5 OFFSET 2;
```

Screenshot:



Query 4: Customers with no orders

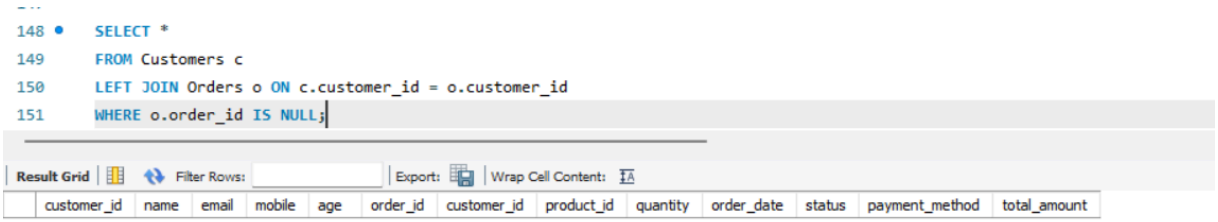
This query finds customers without orders using LEFT JOIN and checking NULL values.

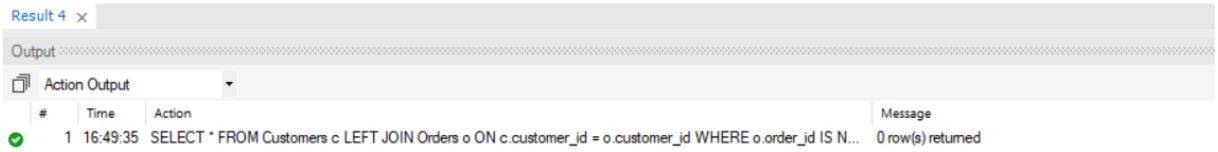
Query:

```
SELECT *
FROM Customers c
LEFT JOIN Orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

Note-All customers have placed orders, so result is empty.

Screenshot:

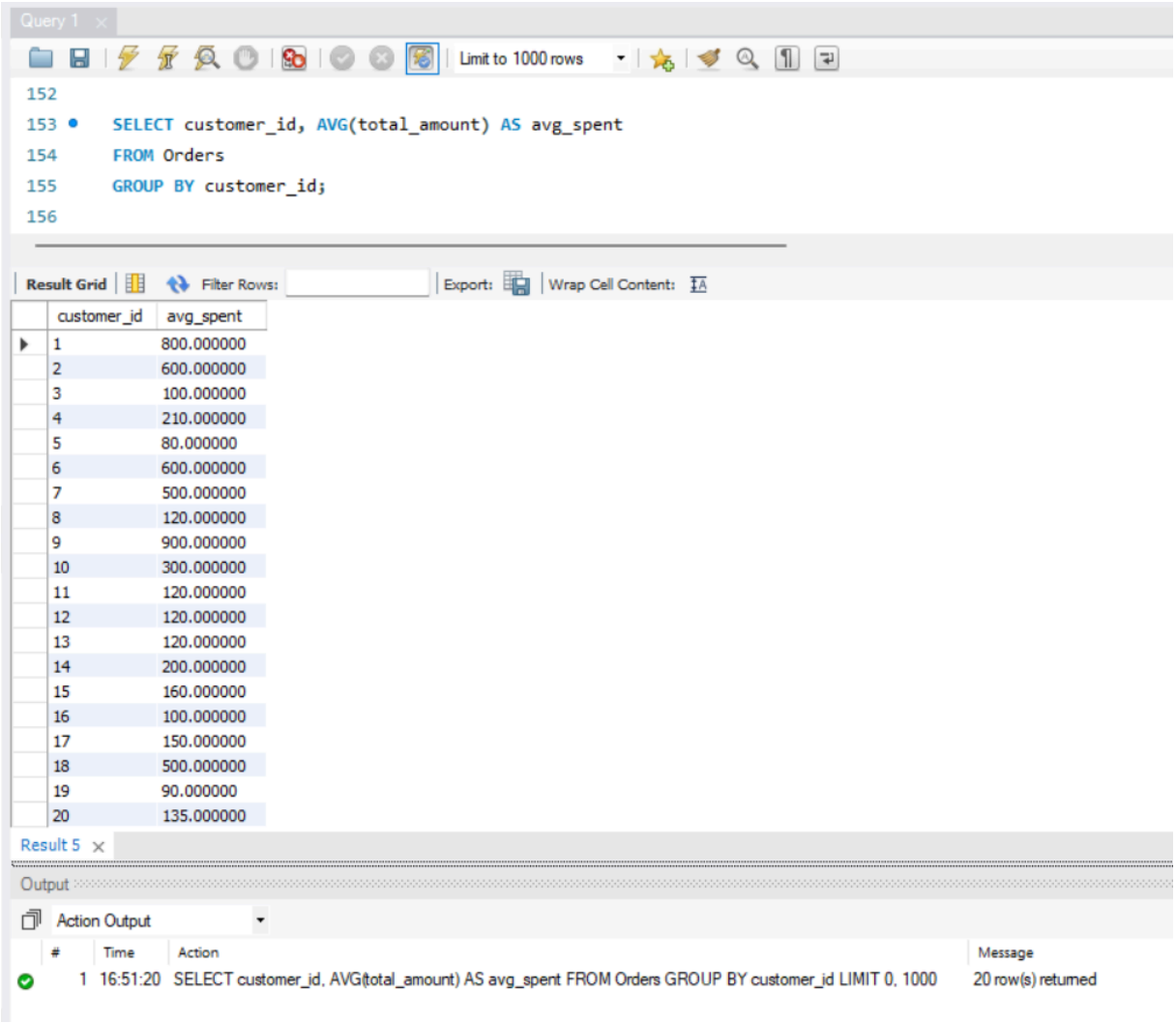




Query 5: Average spending per customer.
This query calculates average spending per customer.

Query:
SELECT customer_id, AVG(total_amount) AS avg_spent
FROM Orders
GROUP BY customer_id;

Screenshot:



customer_id	avg_spent
1	800.000000
2	600.000000
3	100.000000
4	210.000000
5	80.000000
6	600.000000
7	500.000000
8	120.000000
9	900.000000
10	300.000000
11	120.000000
12	120.000000
13	120.000000
14	200.000000
15	160.000000
16	100.000000
17	150.000000
18	500.000000
19	90.000000
20	135.000000

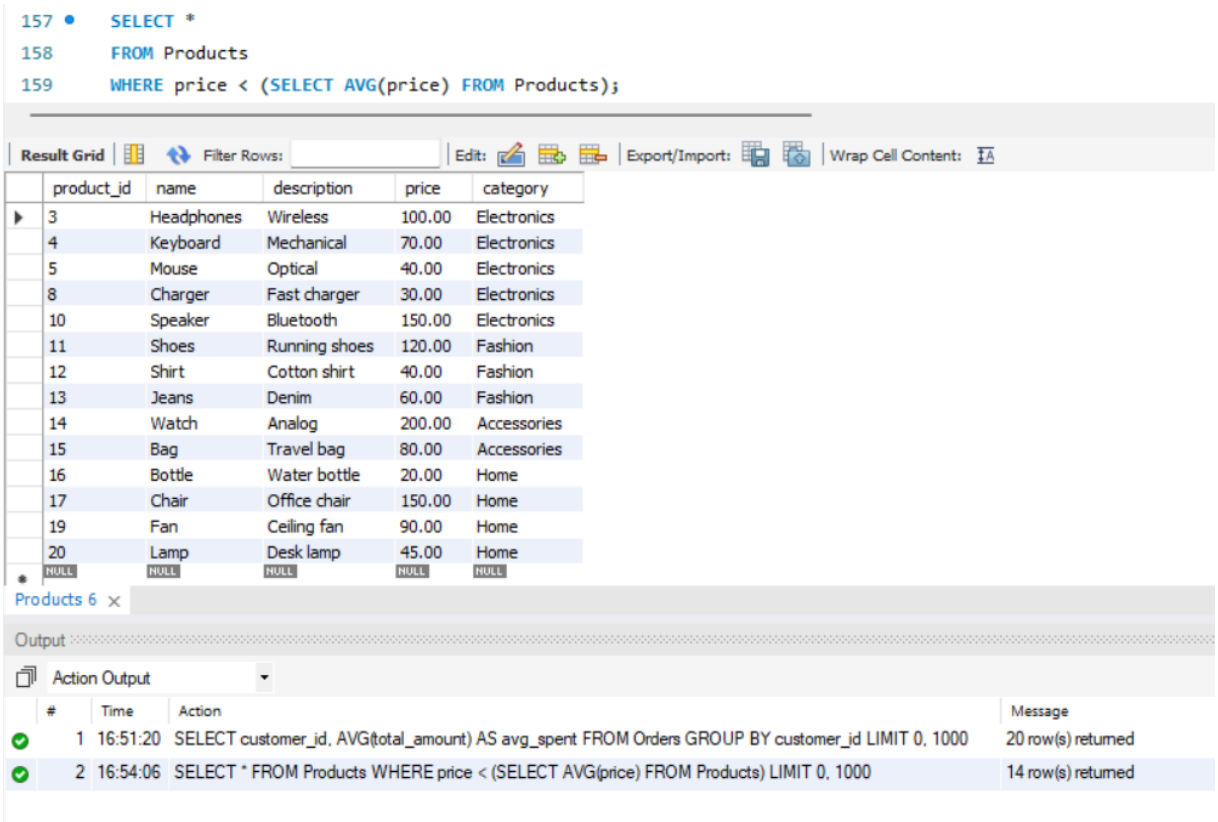
Query 6: Products below average price

This query uses a subquery to compare product price with average price.

Query:

```
SELECT *
FROM Products
WHERE price < (SELECT AVG(price) FROM Products);
```

Screenshot:



```
157 •   SELECT *
158     FROM Products
159     WHERE price < (SELECT AVG(price) FROM Products);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	name	description	price	category
3	Headphones	Wireless	100.00	Electronics
4	Keyboard	Mechanical	70.00	Electronics
5	Mouse	Optical	40.00	Electronics
8	Charger	Fast charger	30.00	Electronics
10	Speaker	Bluetooth	150.00	Electronics
11	Shoes	Running shoes	120.00	Fashion
12	Shirt	Cotton shirt	40.00	Fashion
13	Jeans	Denim	60.00	Fashion
14	Watch	Analog	200.00	Accessories
15	Bag	Travel bag	80.00	Accessories
16	Bottle	Water bottle	20.00	Home
17	Chair	Office chair	150.00	Home
19	Fan	Ceiling fan	90.00	Home
20	Lamp	Desk lamp	45.00	Home
NULL	NULL	NULL	NULL	NULL

Products 6

Output

Action Output

#	Time	Action	Message
1	16:51:20	SELECT customer_id, AVG(total_amount) AS avg_spent FROM Orders GROUP BY customer_id LIMIT 0, 1000	20 row(s) returned
2	16:54:06	SELECT * FROM Products WHERE price < (SELECT AVG(price) FROM Products) LIMIT 0, 1000	14 row(s) returned

Query 7: Total quantity per customer

This query calculates total quantity ordered by each customer.

Query:

```
SELECT customer_id, SUM(quantity) AS total_quantity
FROM Orders
GROUP BY customer_id;
```

Screenshot:

```
SELECT customer_id, SUM(quantity) AS total_quantity
FROM Orders
GROUP BY customer_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	total_quantity
1	1
2	2
3	1
4	3
5	2
6	1
7	2
8	4
9	1
10	2
11	1
12	3
13	2
14	1
15	2
16	5
17	1
18	2
19	1
20	3

Result 8

Output

Action Output

#	Time	Action	Message
1	16:58:44	SELECT customer_id, SUM(quantity) AS total_quantity FROM Orders GROUP BY customer_id LIMIT 0, 1000	20 row(s) returned

Query 8: Orders with customer & product name

This query joins multiple tables to display customer and product names.

Query:

```
SELECT o.order_id, c.name AS customer_name, p.name AS product_name
FROM Orders o
JOIN Customers c ON o.customer_id = c.customer_id
JOIN Products p ON o.product_id = p.product_id;
```

Screenshot:

```
SELECT o.order_id, c.name AS customer_name, p.name AS product_name
FROM Orders o
JOIN Customers c ON o.customer_id = c.customer_id
JOIN Products p ON o.product_id = p.product_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_id	customer_name	product_name
1	Amit Sharma	Laptop
2	Riya Singh	Mobile
3	Rahul Verma	Headphones
4	Sneha Jain	Keyboard
5	Vikas Gupta	Mouse
6	Neha Kapoor	TV
7	Arjun Mehta	Tablet
8	Pooja Sharma	Charger
9	Karan Patel	Camera
10	Simran Kaur	Speaker
11	Manish Yadav	Shoes
12	Anita Das	Shirt
13	Rohit Roy	Jeans
14	Meera Nair	Watch
15	Suresh Kumar	Bag
16	Priya Iyer	Bottle
17	Deepak Singh	Chair
18	Nikita Shah	Table
19	Raj Malhotra	Fan
20	Alok Pandey	Lamp

Result 9

Output

Action Output

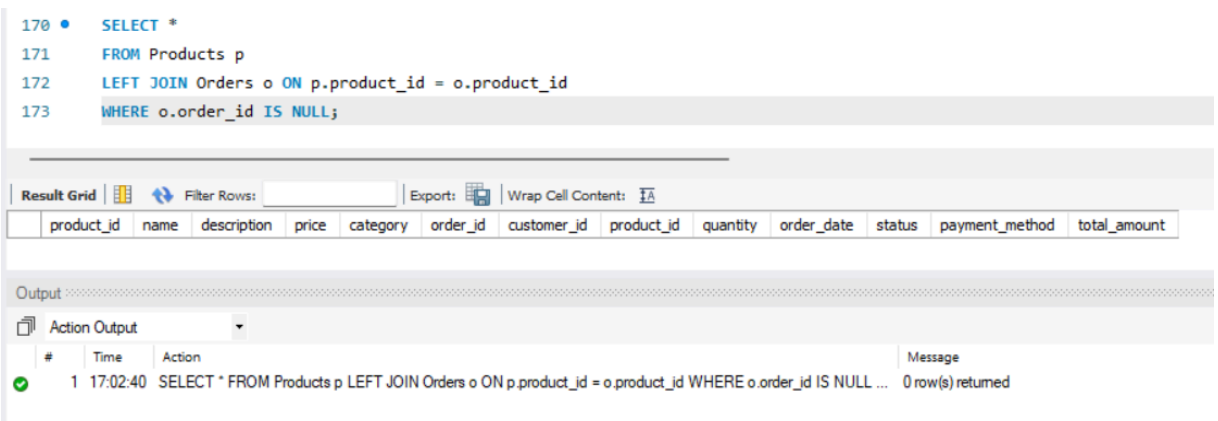
#	Time	Action	Message
1	16:58:44	SELECT customer_id, SUM(quantity) AS total_quantity FROM Orders GROUP BY customer_id LIMIT 0, 1000	20 row(s) returned
2	17:01:00	SELECT o.order_id, c.name AS customer_name, p.name AS product_name FROM Orders o JOIN Customers c ...	20 row(s) returned

Query 9: Products never ordered

This query finds products that have never been ordered.

Query:

```
SELECT *
FROM Products p
LEFT JOIN Orders o ON p.product_id = o.product_id
WHERE o.order_id IS NULL;
```

Note-All customers have placed orders, so result is empty.

Screenshot: